PREETAM MASKE
San Jose, CA | (669)-340-6332 | preetam.a.maske@gmail.com | linkedin/preetam-maske/ | github.com/Preetam3620

EDUCATION

Master of Science, Computer Software Engineering Aug 2024 – May 2026
San Jose State University, San Jose, California
Coursework: Systems Software Engineering, Cloud Technologies, Database Management, Distributed Systems

Bachelor of Engineering, Computer Engineering Jul 2018 – Jul 2022
Savitribai Phule Pune University, India
Coursework: Data Structures & Algorithms, Object Oriented Programming, Software Testing and Quality Assurance

TECHNICAL SKILLS

Languages: Java, Python, JavaScript, TypeScript, SQL, C++, Shell
Frameworks & Tools: React, Node.js, FastAPI, Flask, Apache Kafka, LangGraph, Git, Linux
Databases: MySQL, PostgreSQL, MongoDB, Redis, Supabase, Pinecone
Cloud & Infrastructure: AWS, Docker, Kubernetes, Terraform, CI/CD (CircleCI, Bamboo)
Concepts: OOP, Microservices, Distributed Systems, Test Automation, Agentic AI

WORK EXPERIENCE

Software Engineer, Volkswagen Group Technology Solutions, Pune, India Aug 2022 – Jul 2024

- Architected and delivered the Spare Parts Portal end to end, collaborating with stakeholders across teams to scale React/Node.js framework to 10,000+ users, cutting revision cycles 30%.
- Diagnosed SAP posting bottlenecks under peak load and designed a Kafka pipeline that decoupled OCR, validation, and posting stages, lowering processing latency 40% under load testing.
- Tracked latency telemetry to pinpoint slow MySQL queries in production microservices, then added Redis caching and targeted indexes that trimmed P95 response time 35%.
- Implemented a computer vision OCR pipeline for invoice processing, partnering with finance stakeholders to validate accuracy and reduce manual entry 60%.
- Developed test strategies and automation frameworks for a Bamboo CI/CD pipeline, designing integration, performance, and security test cases that shortened release cycle time 50%.
- Hardened production Flask services with IAM policies, security groups, and automated EC2 health checks, keeping the platform stable with zero security incidents over 2 years.

Software Developer Intern, Simba Developers Jan 2021 – Jul 2021

- Delivered a real-time sensor monitoring dashboard in Node.js, instrumenting live device telemetry at sub-second latency that decreased manual operations reporting effort 70%.
- Diagnosed rendering bottlenecks in a Flutter/Dart app through systematic UI profiling, then optimized paint cycles to cut input-to-render latency 25% on Android and iOS.
- Standardized API contracts and led pre-merge code reviews across a 6-person team, catching integration defects early and lowering post-release bug reports 40%.
- Self-initiated SQL vs. NoSQL architecture analysis for a home automation backend; the team adopted the recommendation, saving 2 weeks of integration work.

PROJECTS

CloudKraft: Cloud Workflow Orchestration Tool | FastAPI, Terraform, LangGraph, AWS Jan 2026 – May 2026

- Orchestrated full Terraform deployment lifecycle (plan, apply, destroy) with non-blocking thread executors and live log streaming, enabling infrastructure provisioning across 20+ AWS resource types.
- Built dual-mode Terraform validation, real CLI checks plus a regex-based security fallback, shrinking validation latency from 30s to 2s through workspace prewarming.
- Accelerated development using Cursor and Claude Code, critically reviewing and refining AI-generated output while building a multi-provider LangGraph chat layer across Claude, GPT-4o, and Gemini.

BookTable: Restaurant Reservations Platform | FastAPI, PostgreSQL, AWS, Terraform Jan 2025 – Mar 2025

- Developed role-based REST APIs in FastAPI with input validation and idempotent request handling, cutting booking conflict errors 90% across three user tiers.
- Configured CircleCI pipelines with automated unit tests, database migrations, and health checks, reducing manual release effort 60% and enabling same-day rollback.
- Provisioned containerized services on AWS EC2 behind an Application Load Balancer via Terraform, sustaining 10,000 concurrent users with zero-downtime blue-green releases.

AlphaRescue: Emergency Dispatch System | React, Mapbox, Supabase, Fetch.ai Jun 2025 – Sep 2025

- Built a GenAI voice assistant on Gemini that transcribed live distress calls and generated structured incident briefs in under 3 seconds end to end.
- Designed Fetch.ai uAgent network for severity classification and RAG-based facility lookup on Supabase, cutting manual dispatch decision time 70%.
- Created real-time React dashboard streaming live incident reports so dispatchers could monitor and intervene as situations unfolded, using Claude Code to accelerate development under a 24-hour hackathon deadline.

ACHIEVEMENTS

UC Berkeley AI Hackathon 2025 - Best AI Agent Deployment, Awarded for AlphaRescue Jun 2025
ISTE Hack-SPRINT v3.0 2021, a 36-hour hackathon - Business Impact Award Sep 2021